

Prospects of Using Karabakh Karizes for Water Supply (a Case Study in The Jabrayil Region)

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Abstract

Climate change and rising temperatures around the world over the last 50 years have shown that people should use water resources, including groundwater, correctly and efficiently. The fact that many scientists and authorities around the world are now focusing their attention on karri systems shows that karri, an ancient hydrotechnical system, has always been at the service of mankind and will continue to be the most promising land reclamation system in the future. Karizes are nature's most benign system and have no negative impact on ecosystems. They are considered to be the most reliable source of water in arid areas. Considering that today's global warming and natural phenomena on the planet call for humanity to be vigilant and to use nature efficiently, greater attention must be paid to the protection and restoration of karri systems, which can provide pure and clean water without interruption, can be repaired at low cost, and can meet the population's needs for drinking and irrigation water. According to available documentation, there are 98 karizes in Karabakh, mainly in the Agdam region, 71 in the Fuzuli region, 111 in Jabrayil and 52 in the mountainous part of Karabakh. The total drainage capacity of these karizes was 122 million m³/year. The karizes in the Jabrayil region have a drainage capacity of 34.68 million m³ per year. According to the results of the analyzes carried out in the karizes of the study area, no foreign substances, especially heavy metals, were found. This also indicates that the indicators in the study area are within the permissible limits. In order to study the impact of using kariz water as irrigation water in the foothill areas on improving soil fertility and productivity, a complete chemical analysis of kariz water used for drinking and irrigation purposes in the villages of Amirvarly, Fuganly, Garajally, etc. in the Jabrayil region was carried out. The analysis revealed that the content of Br, Sr, Cd, Hg, Pb, Zn, Cr, Cu, Se, Mg and As in Karizes water is many times lower than the norm. This gives reason to say that this Karizes water is suitable for use in everyday life and agriculture.

Keywords:

qanat system, galleria, climate change, water resource, groundwater, underground water channel systems

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1. Introduction

Kariz (a Persian word, or Qanat in Somatic/Arabic languages) is a very common name of the hydraulic system in Asia, specifically in Iran, Afghanistan, and Pakistan [1]. This water system has different names in 27 countries, such as Kanerjing (China), Ain (Saudi Arabia), Falaj/Afaj (Oman and the United Arab Emirates), Qanat Romani (Jordan and Syria), Khetara (Morocco), Galleria (Spain), and Foggara/Fujihara (North Africa) [2].

Karizes are of great importance in Azerbaijan as a source of water and life. They are ancient examples of the irrigation and construction culture of Azerbaijan. They are ancient examples of the irrigation and construction culture of Azerbaijan. Karizes are also given serious importance among the people as monuments, or as springs-monuments. Traces of the history of Azerbaijan are preserved in Karizes. For all this, studying Karizes is of great importance in terms of studying the history and culture of Azerbaijan. Karizes are one of the important research objects of Azerbaijani agricultural science [3,4].

According to the UN National Climate Change Data, during 1991-2000, the air temperature in the Republic of Azerbaijan increased by 0.41 °C, and atmospheric precipitation was 9.9% less than normal. The climate scenarios for 2021-2050 indicate that in the first half of the 21st century, the average increase in air temperature in Azerbaijan will be 0.3 °C per 10 years. Therefore, water resources in Azerbaijan should be used more correctly and efficiently [5,6].

Karizes are used to solve the water problem in 41 countries of the world. Azerbaijan ranks second in the world in terms of the use of Kariz water after countries such as Iran, Afghanistan, Pakistan, and Oman. They are considered the most reliable source of water in arid climate zones. Kariz systems do not have any negative impact on the ecosystems in which they exist and the surrounding areas. Karizes are one of the oldest heritage monuments of the Azerbaijani people. Research data on the history of the culture of using surface and groundwater in Azerbaijan, as a country belonging to the arid climate zone, has been obtained. In addition, information on the classification of

kariz according to their food sources, etc., was obtained [5,7].

According to available materials, there are 98 karizes in Karabakh, mainly in the Aghdam region, 71 in the Fuzuli region, 111 in Jabrayil, and 52 in the mountainous part of Karabakh. The total drainage capacity of these karizes are 122 million m³/year [4,7,8].

Jabrayil region is located in the south of the Republic of Azerbaijan, in the southeast of the Lesser Caucasus Mountains, on the left bank of the Araz River. It borders the Islamic Republic of Iran to the south, Zangilan to the southwest, Gubadli to the west, Khojavend to the north, and Fuzuli to the east. Its area is 1049 km² and its terrain is mainly mountainous (Fig. 1.).



Figure 1: *Jabrayil district*

The territory of the district is completely limited by the river network. Since ancient times, the population's demand for drinking and irrigation water has been met through karizes and springs. The territories liberated from occupation have fertile lands, potential water resources and are capable of making significant contributions to the country's economy. According to the available information, Armenian vandals have brutally destroyed some karizes. Mines, corpses of prisoners, various objects and ammunition have been poured into kariz wells. In this regard, it can be said that monitoring karizes in the liberated territories, working out the scientific basis for their restoration and use, is very relevant at a time when the demand for drinking and irrigation water is increasing. The territories liberated from occupation have fertile lands, water resources, including kariz water resources, and their use can make significant contributions to the country's economy [3,5,8,9].

2. Methods

The Jabrayil district basin is part of the Kur-Araz lowland (Table 1). Groundwater in the plain is distributed in modern alluvial and alluvial-proluvial, Hakara formation, Cretaceous and Jurassic sediments [4,9-11].

Table 1: *Hydrogeological basins in Azerbaijan*

Geostructural regions	Hydrogeological basins
A. Fold Mountains Area of the Greater Caucasus	1. Porous-fractured water basin of the Greater Caucasus; 2. Shamakhi-Gobustan porous-stratal and fractured water basin; 3. Porous-stratal water basin in Absheron Peninsula; 4. Samur-Gusarchay porous-stratal water basin;
B Fold Mountains Area of the Lesser Caucasus	1. Porous-fractured water basin of the Lesser Caucasus; 2. Nakhchivan porous-fractured water basin; 3. Talysh porous-fractured water basin;
C. Kur-Araz Lowland	1. Sheki-Zagatala porous-stratal water basin; 2. Ganja-Gazakh porous-stratal water basin; 3. Karabakh porous-stratal water basin; 4. Mil porous-stratal water basin; 5. Jabrayil porous-stratal water basin; 6. Nakhchivan porous-stratal water basin; 7. Shirvan porous-stratal water basin; 8. Mugan-Salyan porous-stratal water basin; 9. Lankaran porous-stratal water basin; 10. Porous-stratal water basin in Neogenic foothills of Jeiranchel; 11. Porous-stratal water basin in Neogenic foothills of Ajinour.

In the Jabrayil foothill plain, groundwater in the headwater of river inflow cones creates a single unpressurized aquifer. Intensive absorption of atmospheric sediments and surface water is taking place in this zone. In the central and peripheral parts of the intake cones, the single aquifer is directly divided into two unpressurized (ground) and one pressurized (artesian) aquifers with the appearance of clay-

clay layers or clay-filled rocks in the lithological section. The groundwater horizon (unpressurized) is associated with alluvial sediments of river valleys and sediments of the Hakara group beneath them. Main average geological parameters of the aquifers are given in Table 2.

Table 2. *Characteristics of aquifers along Jabrayil porous-stratal water basin of foothill plain*

Hydro-geological region	Aquifer	Occurrence depth of aquifer, meters	Piezometric level in relation to ground surface, meters	Thickness of aquiferous rocks, meters	Well output rate, l/s	Specific yield l/s.meter	Permeability of aquiferous rocks, m/d
Jabrayil	Unconfined (Alluvial)	-	0,70-30	10,80-119,1	0,14-9,10	0,01-3,1	3,0-24,20
	Unconfined (Hakara)	-	20-81	23-210	0,07-18	0,01-1,80	0,01-6,80
	Confined	200,0-300,0	+17-81	15-57	0,21-5,70	0,003-0,90	0,01-5,50

The aquifers in Jabrayil district have a capacity of 34.68 million m³ per year. There are a large number of aquifers in the Jabrayilchay, Gozluchay and Kondalanchay basins, the total flow of which is about 10-15 l/sec.

There are relatively few aquifers in the Hakara river basin, and their flow does not exceed 2-4 l/sec. There are no aquifers in the Okchuchay and Khachinchay basins. The most abundant aquifer is observed in alluvial-proluvial sediments, which are also attributed to the Hakara, Jabrayilchay and Kondalanchay basins (Table 3).

The aquifer complex of the Hakara formation is spread throughout the entire area of the Jabrayil foothill plain. This aquifer complex is composed of gravel and pebbles filled with sand and clay. This complex is organized according to plans and sections from a large number of aquifer horizons, which are separated from each other by clay layers.

Table 3: *Characteristics of some hydrogeological parameters of aquifers*

River basin	Filtration coefficient, K _f , m/sec	Water permeability K _m , m ² /day	Consumption per unit width of the horizon, l/sec.m
Khachinchay	5-12	80-120	0,25-0,36
Hakara	1,2-12,8	20-200	0,17-1,14
Jabrayilchay	7,8-24,2	180-650	0,4-3,0
Kondalanchay	5-15	300-1800	0,6-1,8
Okchuchay	0,3-3,7	<100	0,14-0,17

The classical model is used for the technology of excavation of karizes in the Jabrayil region (Fig. 1.)

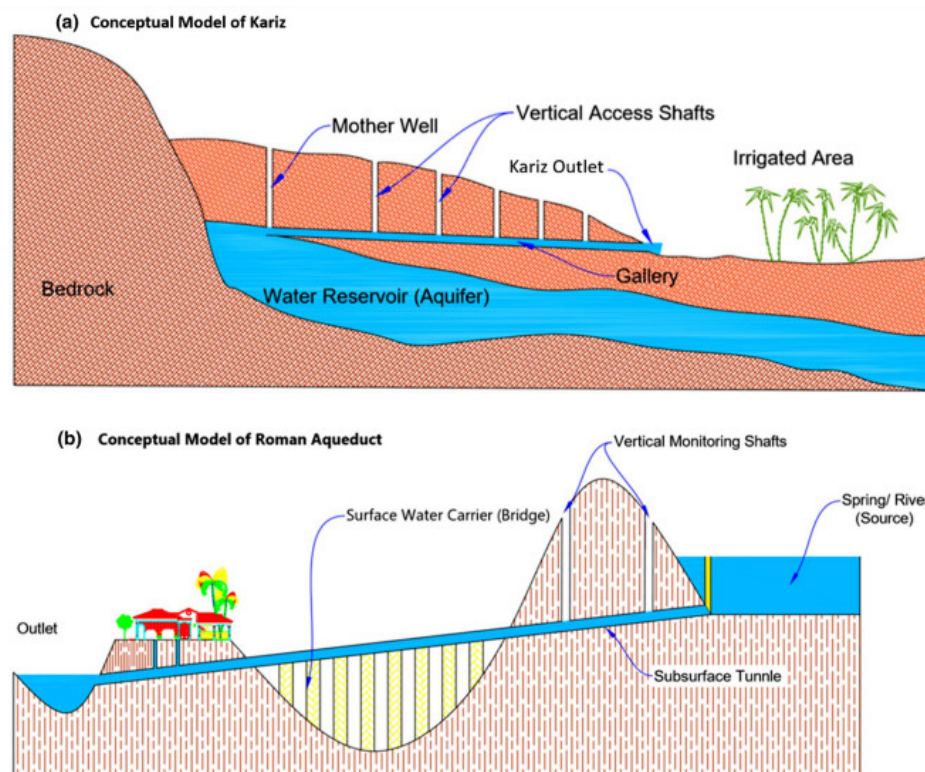


Figure 1: *Conceptual models of a typical Kariz (a) and Roman Aqueduct system (b)*

In the study area, restoration works are being completed in 5 fields. Some parameters of these karizes have been determined and are shown in the table below (Table 4).

According to their chemical composition, groundwater in the Jabrayil foothill plain is hydrocarbonate in continental sediments, rarely hydrocarbonate-sulfate, calcium, or calcium-magnesium. Hydrocarbonate-sulfate

water are often found in the Okchuchay valley, and sulfate water are found in the Injehay-Chakhmakhay interfluvial areas and along the Jabrayilchay and Araz rivulets (from the Chaylaq river to the Kondalanchay). Calcium predominates among cations. Sodium predominates in areas where sulfate water spread. Here, the degree of

mineralization of water is found to be up to 2.5 g/l. In general, from the foothills to the near Araz area, hydrocarbonate water pass into sulfate water, and in some places into chloride water. In this regard, their degree of mineralization increases.

Table 4: Parameters of restored kariz in Jabrayil district

Name of Kariz	Location	Length, m	Vertical shafts	Water consumption l/sec
Gulgasm	Jabrayil, Horovlu village	553.5	14	4
Shigi	Jabrayil, Horovlu village	481.4	22	4
Khalifa Valley	Jabrayil, Horovlu village	477.5	23	3
Asgar kariz	Jabrayil, Horovlu village	500	11	15
Orta kariz	Jabrayil, Horovlu village	450	3	-

In the Jabrayil foothill plain, water complexes of the Jurassic and Cretaceous systems lie at great depths. They are located in volcanic sediments of marine origin. In terms of lithological composition, they consist mostly of thick limestones and sandstones. The water content of the rocks is low. Their feeding is located in the mountainous parts of the region. Accordingly, they lie in homogeneous sediments that do not erode the bottom of continental sediments, and there is no interaction with old rocks.

In the deluvial zone of the study area, groundwater has a low mineralization since it is in relatively deeper layers. In the plain areas, since their depth is close to the surface, the degree of mineralization is also high. In this regard, as a result of the influence of hydrogeological factors, on the one hand, hydromorphic soils, and then, due to the violation of the regime, conditions for gradual waterlogging and salinization of soils have been created. After starting monitoring work on the karizes in the liberated and demined areas, water samples were taken from the water of the karizes in the study area for the purpose of conducting physical and chemical analyses and analyzed in the relevant laboratories. The results of the physical and chemical analysis of water samples taken from the Dash Veysalli village - Kichik kariz, Kavdar village - Agalarbay kariz, Charaken village - Taveka kariz, Horovlu village - Shigi kariz, Khalifa valley kariz, Askar kariz, Orta kariz, and Guzey kariz of the Gujjak village of the Jabrayil region showed that this water are completely suitable for drinking and irrigation [3,12-14]. Thus, the

turbidity of the water in the karizes varies between 0.21-2.5 mg/l and does not exceed the permissible concentration limit of 2.6 mg/l. Only the turbidity indicator of the water sample taken from the Dashveysalli village Kichik kariz was slightly higher than the norm - 2.71 mg/l. This is related to the urban development and restoration works carried out by some Turkish companies in that area. Thus, when the depth of the wells is less than 4-5 m, collapses occur as a result of the impact of heavy equipment moving over the tunnel. Thus, the results of the conducted studies showed that the water of the wells is fully suitable for use in drinking water supply (Table 5) [3,6,11,15,16].

Large-scale battles were fought in the territory of the Jabrayil region. For about 30 years, various explosions, mining and other processes were carried out in this area. Today, part of these areas has been cleared of mines and military waste and put into use, while most of them are still kept as a prohibited combat zone [3,6,12-14]. Appropriate monitoring work was carried out in all the wells inspected in the areas cleared of mines during the research period. The impact of war crimes during the occupation was everywhere and played an important role in changing the soil water regime. During the war, artillery shells, mines and dug trenches caused the collapse of the entire structure of the area. Thus, the analysis of the well water taken from the combat zone was analyzed in the relevant laboratory of the Institute of Radiation Problems. The analysis results are presented in tabular and explanatory form.

Table 5: Analysis results of samples taken from the water of the Karizes in the Jabrayil region

№	Point description	pH	COND msm/cm	TDS mg/L	SAL %	DO %	DO mg/L	T °C	Rad μSv/h
1	Horovlu village Askar karizi	8,06	644	322	0,03	98,4	8,33	21,1	0,001
2	Horovlu village, Orta kariz	7,88	302	371	0,04	95,6	8,48	21,2	0,001
3	District center Chinar Karizi	7,80	407	203	0,02	97,9	8,46	21,1	0,001

Note: Analyses were conducted in the appropriate laboratory of the Institute of Radiation Problems of ANAS. COND – Electrical conductivity, TDS – total dissolved solids, SAL – salinity, DO – dissolved oxygen (percentage and mg/l), Rad – denotes radiation.

Table 5 shows that according to the analysis results, no foreign substances, especially heavy metals, were encountered. This also indicates that the indicators in the

studied area are within the permissible limits. In order to study the impact of using Kariz water as irrigation water in the foothill zones on improving soil fertility and

productivity, a complete chemical analysis of Kariz water used for drinking and irrigation purposes in the villages of Amirvarli, Fuganli, Garajalli, etc. of Jabrayil district was carried out. The analysis results are given in Table 6.

Table 6: Complete chemical analysis of Karizes water

Water composition	Amirvarli	Binadarasi	Fuganli	Niyazgulular (spring)	Karajalli	Normal
hardness	8,77	3,87	14,54	8,28	5,30	---
pH	7,80	8,02	7,99	8,01	8,15	---
Na	98,42	1,7711	16,92	49,38	2,19	200
K	2,16	0,36	55,45	30,44	15,93	50
Ca	120,24	66,40	275,22	141,49	96,64	180
Mg	33,11	6,70	9,45	14,52	5,73	40
Al	0,029	0,031	0,039	0,048	0,042	0,5
Fe	0,15	0,14	0,11	0,13	0,12	<03
Sulfate	42,84	2,15	21,06	12,96	2,19	<<500
Carbonate	326,14	120,90	495,50	323,24	170,34	---
Chloride	61,74	1,34	9,54	19,32	3,14	<350
Si	16,17	2,52	17,37	8,52	4,38	10
Br	0,02	0,03	0,04	0,05	0,07	0,2
Sr	0,84	0,91	1,12	0,95	0,87	7
Cd	0,0004	0,0005	0,0003	0,0003	0,0004	0,001
Hg	0,0001	0,0002	0,0001	0,0002	0,0001	0,0005
Pb	0,002	0,001	0,003	0,001	0,003	0,03
Zn	0,005	0,009	0,008	0,007	0,004	1
Cr	0,009	0,008	0,012	0,011	0,008	0,1
Cu	0,009	0,016	0,017	0,024	0,028	1
Se	0,0005	0,0002	0,0003	0,0003	0,0005	0,01
Mn	0,012	0,014	0,021	0,012	0,011	0,1
As	0,002	0,003	0,004	0,002	0,005	0,05
Dry residue	701,89	203,30	901,88	601,10	301,70	1000 (1500)
Coordinate Latitude	39.296451	39.452005	39.351899	39.257520	39.470299	
Coordinate Longitude	47.076540	47.005217	47.088041	46.955593	47.088776	
Altitude (m)	154	807	384	328	613	

Note: The analyses were conducted in the relevant laboratory of the Institute of Geology and Geophysics of the Academy of Sciences of the Republic of Azerbaijan.

Table 6 shows that the hardness of the Karizes water is normal, pH is between 7.8-8.02, while the permissible limit of Sodium is 200 mg/l, it is between 1.7711-98.42 (the amount of Na increases towards the Araz River), while the norm of Potassium (K) is 50 mg/l, it was 2.16 in the Amirvarli village, 0.36 in the Binadarasi area, and 55.45 mg/l only in the Fuganli village.

While the permissible limit of Calcium (Ca) in the water is 180 mg/l, it was 120.24 in the Amirvarli village, and 66.40 mg/l in the Binadarasi village.

Although the amount of Magnesium (Mg) is 40 mg/l, it is 33.11 in the Amirvarli village, and 6.70 mg/l in the Binadarasi village.

The amount of aluminum (Al) and iron (Fe) is much lower than the norm.

While the norm for sulfates is 500 mg/l, it is 42.84 in Amirvarli village, and 2.15 mg/l in Binaderesi. Carbonates are equal to 495.5 - 120.90 mg/l across the area.

While the norm for chlorides is lower than 350, it is only 61.74 in Amirvarli village, and 1.34 mg/l in Binadarasi village.

The amount of silicon is up to 10 mg/l. This value is 16.17 in Amirvarli, and 2.52 mg/l in Binadarasi.

Naturally, the change of silicon is also regular. In general, the content of Br, Sr, Cd, Hg, Pb, Zn, Cr, Cu, Se, Mn and As in Karizes water is many times lower than the norm.

3. Results

As a result of the conducted research, it was determined that the prospects for using karizes water in Jabrayil district are very high. Thus, salinization of lands near crops can be prevented to some extent by extracting kariz water. Kariz water can be used as fresh water in everyday life. Karizes are also considered important in the improvement of deserts. Using kariz water in irrigation is also of great importance.

Finally, in some provinces where karizes exist, the preservation of ancient and medieval buildings as historical and architectural monuments can turn these places into attractive tourist attractions.

CONFLICT OF INTEREST

Authors declare that they do not have any conflict of interest.

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